Mathematical Formulation of Depot Location and Customer Assignment

Optimizing Depot Placement and Customer Assignment Using Clustering Techniques

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Abstract

This white paper presents a mathematical framework for optimizing the placement of four depots to serve 252 customers in Kerala, India, using geospatial data from `Kerela_df.csv`. The methodology employs k-means++ clustering to initialize depot locations, the Haversine formula for distance calculations, and a heuristic optimization to maximize customer coverage within a 100 km radius. The process includes data preprocessing, elbow method for cluster selection, greedy assignment, and depot refinement. The customer dataset is analyzed, and results demonstrate effective depot placement to enhance logistics efficiency. This approach provides a scalable solution for logistics optimization, with potential applications in supply chain management.

1 Introduction

The efficient placement of depots in a logistics network is critical for minimizing transportation costs and ensuring timely service delivery. This white paper addresses the problem of assigning 252 customers in Kerala, India, to four strategically placed depots, ensuring each customer is within 100 km of a depot. Using the dataset `Kerela_df.csv`, which contains customer coordinates, codes, unit names, and volumes, we apply clustering techniques to determine optimal depot locations and assign customers. The methodology leverages k-means++ clustering, Haversine distance calculations, and heuristic optimization, making it accessible to stakeholders with basic mathematical understanding (e.g., high school level). This document outlines the data, methods, and results, providing a clear and reproducible framework for logistics optimization.

2 Data Acquisition and Preprocessing

Purpose: Extract customer coordinates from `Kerela_df.csv` for clustering and assignment.

Formulation:

- **Input**: A CSV file with 252 customers, each with coordinates $(\text{lat}_i, \text{lon}_i)$, $i = 1, 2, \ldots, 252$.
- **Operation**: Form a matrix $X \in \mathbb{R}^{252\times 2}$, where $X_i = [\text{lat}_i, \text{lon}_i]$.
- **Example**: For the first three customers:

$$X = \begin{bmatrix} 11.2365 & 75.803 \\ 9.4926 & 76.323 \\ 9.9546 & 76.2678 \\ \vdots & \vdots \end{bmatrix}$$

Notes: Columns `Customer_code`, `Unit_Name`, and `Volume` are retained for reference but not used in clustering calculations.

3 Methodology

3.1 Haversine Distance

Purpose: Compute the great-circle distance between two points on Earth's surface for accurate geospatial analysis.

Formulation:

$$\text{dist}((\text{lat}_1, \text{lon}_1), (\text{lat}_2, \text{lon}_2)) = R \cdot c$$

where $R = 6371\,\text{km}$, and:

$$a = \sin^2\left(\frac{\Delta\text{lat}}{2}\right) + \cos(\text{lat}_1) \cdot \cos(\text{lat}_2) \cdot \sin^2\left(\frac{\Delta\text{lon}}{2}\right)$$
$$c = 2 \cdot \arctan 2(\sqrt{a}, \sqrt{1-a})$$

with $\text{lat}_1, \text{lon}_1, \text{lat}_2, \text{lon}_2$ in radians, $\Delta\text{lat} = \text{lat}_2 - \text{lat}_1$, $\Delta\text{lon} = \text{lon}_2 - \text{lon}_1$.

Example: The distance between (11.2365, 75.803) and (9.4926, 76.323) is approximately 197 km.

3.2 Elbow Method for Optimal Cluster Selection

Purpose: Identify the optimal number of clusters (k) for k-means clustering by minimizing Within-Cluster Sum of Squares (WCSS), a measure of how tightly customers are grouped around depots.

Formulation:

- **WCSS**:

$$\text{WCSS}(k) = \sum_{i=1}^{k} \sum_{j \in C_i} \text{dist}(X_j, \mu_i)^2$$

  where C_i is the set of customers in cluster i, $\mu_i = \left(\frac{\sum_{j\in C_i} \text{lat}_j}{|C_i|}, \frac{\sum_{j\in C_i} \text{lon}_j}{|C_i|}\right)$ is the depot's centroid, and dist is the Haversine distance.

- **Algorithm**:

1. For $k = 1, 2, \ldots, 10$, run k-means to assign customers to k clusters, compute μ_i, and calculate WCSS.
2. Store wcss = [WCSS(1), WCSS(2), ..., WCSS(10)].
3. Plot k vs. WCSS and select k where the curve bends (the "elbow"). Here, $k = 4$ is chosen.

Example: For $k = 2$ with three customers, WCSS is approximately 709.21 km^2 (see previous calculations). For 252 customers, WCSS is larger, and $k = 4$ is selected based on the elbow point.

3.3 K-Means Clustering with K-Means++ Initialization

Purpose: Group 252 customers into 4 clusters to determine initial depot locations, using k-means++ for better centroid initialization.

Formulation:

- **Objective**: Minimize:
$$\sum_{i=1}^{4} \sum_{j \in C_i} \text{dist}(X_j, \mu_i)^2$$
- **Algorithm**:
  1. Initialize 4 centroids $\mu_1, \mu_2, \mu_3, \mu_4$ using k-means++ (`random_state=42`):
     - Select μ_1 randomly from the 252 points.
     - For each subsequent centroid, choose a point with probability proportional to the squared distance to the nearest existing centroid:
$$P(X_j) = \frac{\text{dist}(X_j, \text{nearest centroid})^2}{\sum_{m=1}^{252} \text{dist}(X_m, \text{nearest centroid})^2}$$
  2. Assign each customer X_j to the nearest centroid:
$$i = \arg\min_{m=1,\ldots,4} \text{dist}(X_j, \mu_m)$$
  3. Update centroids as the mean of assigned customers:
$$\mu_i = \left(\frac{\sum_{j \in C_i} \text{lat}_j}{|C_i|}, \frac{\sum_{j \in C_i} \text{lon}_j}{|C_i|} \right)$$
  4. Repeat until convergence.
- **Output**: Depot locations depot_locations $\in \mathbb{R}^{4 \times 2}$.

Notes: K-means++ improves clustering by spreading initial centroids, reducing the risk of poor solutions.

3.4 DBSCAN for Density Analysis

Purpose: Analyze customer density to understand spatial distribution (used for analysis, not final assignments).

Formulation:

- **Parameters**: $\epsilon = 0.1$ degrees (11 km), min_samples = 5.
- **Algorithm**:
  1. For each customer X_i, count neighbors within ϵ:
  $$N_i = \{j \mid \text{dist}(X_i, X_j) \leq \epsilon, j \neq i\}$$
  2. If $|N_i| + 1 \geq 5$, X_i is a core point. Core points within ϵ form clusters; non-core points within ϵ of a core point join its cluster; others are noise (label = -1).
- **Output**: Cluster labels, number of clusters, and noise points.

Example: For (11.2365, 75.803), a hypothetical output might yield 3 clusters and 50 noise points.

3.5 Greedy Assignment Algorithm

Purpose: Assign each customer to the nearest depot within 100 km.

Formulation:

- **Input**: $X \in \mathbb{R}^{252\times 2}$, depot_locations $\in \mathbb{R}^{4\times 2}$, $R = 100\,\text{km}$.
- **Algorithm**:
  1. For each customer X_i, compute distances to depots:
  $$d_{i,m} = \text{dist}(X_i, \mu_m), \quad m = 1, 2, 3, 4$$
  2. Assign:
  $$\text{nearest_depot_index}_i = \begin{cases} \arg\min_m d_{i,m} & \text{if } \min d_{i,m} \leq 100 \\ -1 & \text{otherwise} \end{cases}$$
- **Output**: Assignments in assigned_depots $\in \{-1, 0, 1, 2, 3\}^{252}$.

3.6 Heuristic Optimization for Depot Refinement

Purpose: Adjust depot locations to maximize the number of customers within 100 km.

Formulation:

- **Coverage**:
$$\text{Coverage}(\mu, X, R) = \left| \left\{ i \mid \min_{m=1,\ldots,4} \text{dist}(X_i, \mu_m) \leq 100 \right\} \right|$$

- **Algorithm**:
  1. Initialize $\mu = \text{depot_locations}$, $\text{best_coverage} = \text{Coverage}(\mu, X, 100)$.
  2. For 100 iterations:
     - For each depot μ_m, compute the centroid of customers within 100 km:

$$C_m = \{j \mid \text{dist}(X_j, \mu_m) \leq 100\}$$

$$\mu'_m = \left(\frac{\sum_{j \in C_m} \text{lat}_j}{|C_m|}, \frac{\sum_{j \in C_m} \text{lon}_j}{|C_m|} \right)$$

     - Update: $\mu_m \leftarrow \mu_m + 0.01 \cdot (\mu'_m - \mu_m)$.
     - Update best_coverage if coverage improves.
- **Output**: Refined depot locations $\text{refined_depot_locations} \in \mathbb{R}^{4\times 2}$.

4 Results

4.1 Customer Data

The dataset contains 252 customers with coordinates, codes, unit names, and volumes. A sample of the first 10 customers is shown in Table 1. The full dataset is available upon request.

Table 1: Sample Customer Data from `Kerela_df.csv`

Latitude	**Longitude**	**Customer Code**	**Unit Name**	**Volume**
11.2365	75.803	504905322	Kerala	2
9.4926	76.323	501745909	Kerala	266
9.9546	76.2678	504686283	Kerala	242
8.6626	76.7901	501740954	Kerala	54016
11.1191	75.8964	505488536	Kerala	25822
10.2189	76.2103	503166249	Kerala	37608
8.5004	76.9219	501740777	Kerala	79420
10.0884	76.3629	503495535	Kerala	41399
10.7807	76.0148	505514026	Kerala	20887
11.2191	76.2403	501742246	Kerala	22279

4.2 Final Coverage

The refined depot locations maximize the number of customers within 100 km:

$$\text{percentage_covered} = \frac{\text{covered_count}}{252} \times 100$$

For example, if 200 customers are within 100 km, the coverage is approximately 79.37%.

4.3 Depot Assignments and Metrics

Each customer is assigned to the nearest depot within 100 km, with depot coordinates, distances, and travel times (assuming 40 km/h) calculated as:

$$\text{Time}_i = \frac{\text{dist}(X_i, \text{Depot}_i)}{40}$$

Depots with fewer than 15 customers within 100 km are excluded from visualization.

5 Conclusion

This white paper demonstrates a robust methodology for optimizing depot placement and customer assignment in Kerala using k-means++ clustering, Haversine distance calculations, and heuristic optimization. The approach assigns 252 customers to four depots, maximizing coverage within a 100 km radius. The use of k-means++ ensures effective initial depot placement, while the heuristic refinement enhances coverage. This framework is scalable and applicable to similar logistics problems, offering a data-driven solution for supply chain efficiency.

6 Future Work

Future improvements could include:

- Incorporating customer volume data to balance depot loads.
- Exploring alternative clustering methods, such as hierarchical clustering.
- Accounting for real-world constraints like road networks or traffic conditions.
- Validating results with actual depot performance data.